Design of a 400 Gb/s Silicon Photonics Receiver

For IEEE 802.3 DR4 Intra-DC Interconnects

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Motivation and Targets

- Intra-datacenter PAM-4 receivers need **high responsivity**, **very low dark current**, and **bandwidth above** 100 GHz.
- A **Ge-on-Si vertical n-i-p photodiode** offers a strong fit with standard SOI photonics.
- The chapter benchmarks the design against Yang Shi et al., *Photonics Research* 12(1), 2024.

Metric	Target
Responsivity	≥ 0.95 A/W
Dark current	≤ 1.3 nA
Bandwidth	≥ 103 GHz
Detectivity	$\geq 2.95 \times 10^{10}$ cm$\sqrt{\text{Hz}}$/W

Device Architecture

- Vertical **n-i-p** stack on a $220\,\mathrm{nm}$ **SOI** platform.
- $50\,\mathrm{nm}$ **n++ Ge** top contact layer.
- $350\,\mathrm{nm}$ **intrinsic Ge** absorber.
- **p++ Si** platform for the anode path.
- **U-shaped electrode** reduces the RC penalty while preserving carrier collection.

Layer	Thickness
n++ Ge	$50\,\mathrm{nm}$
i-Ge	$350\,\mathrm{nm}$
p++ Si	$220\,\mathrm{nm}$
BOX	$2\,\mathrm{\mu m}$

Optical Simulation Flow

- **Lumerical FDTD** models the taper-coupled optical absorption at $1310\,\text{nm}$.
- The simulation exports **optical generation rate** into CHARGE.
- Ge length is optimized to reach high absorption with minimum footprint.

Key optical design points:

- TE-driven O-band operation
- Adiabatic spot-size converter ($500\,\text{nm} \to 5\,\mu\text{m}$ over $40\,\mu\text{m}$)
- 3D generation-rate export: $G = P_{\text{abs}}/(h\nu)$

figures/opt_generation_rate_ma

Electrical Simulation Flow

- **Lumerical DEVICE CHARGE** solves carrier transport under reverse bias.
- SRH recombination enabled with $\tau_n = \tau_p$ (calibrated to match dark current target).
- Fine mesh override: $20\,\text{nm}$ max edge length over the Ge active region.
- Extracted quantities include:
  - Dark current and illuminated current
  - Band diagram, carrier density
  - Electric field and potential
- Bandwidth estimated from **transit-time + RC** limits.

Dark Current Result

- All parameters flow from solvers — no hardcoded paper values.
- Low dark current is essential for:
  - Receiver sensitivity
  - Noise performance
  - Robust PAM-4 eye opening

figures/dark_iv.png

Internal Device Physics

figures/phys_band_diagram.png

figures/phys_carrier_density.png

Band alignment, carrier density, and depletion shaping near $V = -1\,\mathrm{V}$ explain the tradeoff between responsivity, dark current, and speed.

Electric Field and Bandwidth

figures/phys_field_potential.png

figures/freq_response_model.png

- Transit-time and RC contributions are combined into the bandwidth model.
- All parasitic parameters (R_s, C_j, L_p, C_p) derived from geometry.

System-Level Receiver View

figures/system_transfer_function.png

figures/system_photocurrent_eye.p

- The repo extends the device into a **MATLAB PAM-4 receiver model** at 53.125 GBd.
- All device parameters flow from the CML `.mat` generated by device-level postprocessing.

PAM-4 Signal Quality

figures/system_pam4_histogram.png

- Matched-filter output shows **4 distinct PAM-4 levels**.
- Decision thresholds separate the symbol clusters.
- SNR, BER, and SER are computed from the simulation.
- The PD can now be studied both as a device and as a circuit block.

INTERCONNECT-Ready Compact Model

- The repo exports a **Lumerical INTERCONNECT-ready** photodetector data package.
- Generated artifacts include:
  - Responsivity table (frequency-dependent)
  - Bandwidth table (bias-dependent)
  - Dark-current table
  - Compact-model `.mat` source data
  - Equivalent-circuit parameters CSV

`figures/system_interconnect_compact_model.png`

Data Pipeline Overview

`FDTD .lsf` → `fdtd_results.mat` → `CHARGE .lsf` → `charge_results.mat` → `postprocess.m` → `CML .mat` → `system-level`

All parameters flow from Lumerical solvers → MAT files → MATLAB postprocess → CML export → system-level scripts. **Nothing is hardcoded in the pipeline.**

Final Takeaways

- The chapter demonstrates a **Ge-on-Si O-band photodiode** targeting published high-speed metrics.
- The workflow is consistent across:
  - **Device-level FDTD** — optical absorption, generation rate
  - **Device-level CHARGE** — dark current, responsivity, band structure
  - **Device-level postprocess** — CML dataset, all derived parameters
  - **System-level MATLAB** — PAM-4 BER, eye diagrams, histograms
  - **INTERCONNECT-ready export** — circuit-level integration
- **Zero hardcoded paper values** — all metrics from solvers and geometry.